

Entropy Stream Analysis

Deviation from Baseline — Per-Metric Report

Run Parameters

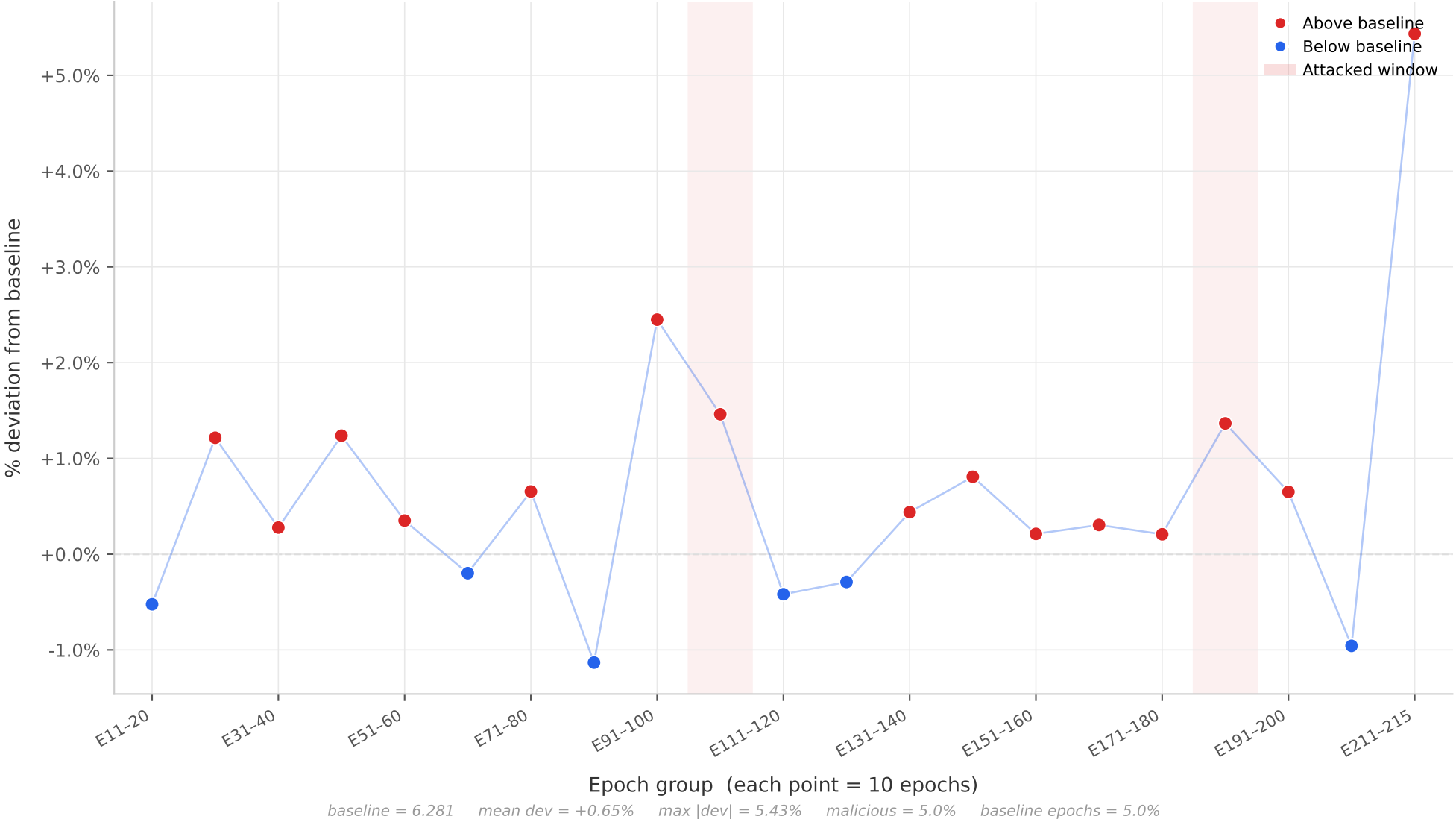
Dataset	epochs_5.0_5.0_sparse.json
Directory	CAIDA_Dataset/txtD280/epochs/
Timestamp	2026-02-19T18:49:27.266117
Total epochs	215
Baseline epochs	10 (5.0%)
Attack epochs	205
Malicious traffic	5.0%
Epoch group size	10 epochs per point
Metrics plotted	16
Mode	SPARSE
Attack probability	0.3
Windows total	21
Windows attacked	2
Windows clean	19

Baseline Values

Src IP Entropy	6.281
Dst IP Entropy	6.09
Src Port Entropy	1.991
Dst Port Entropy	6.34
Src IP Cardinality	5022
Dst IP Cardinality	5005
Src Port Cardinality	4288
Dst Port Cardinality	5020
Src IP Uniformity	0.01556
Dst IP Uniformity	0.01364
Src/Dst Ratio	1.004
Total Flows	5056
Unique Src IPs	3567
Unique Dst IPs	2795
Unique Src Ports	2317
Unique Dst Ports	3816

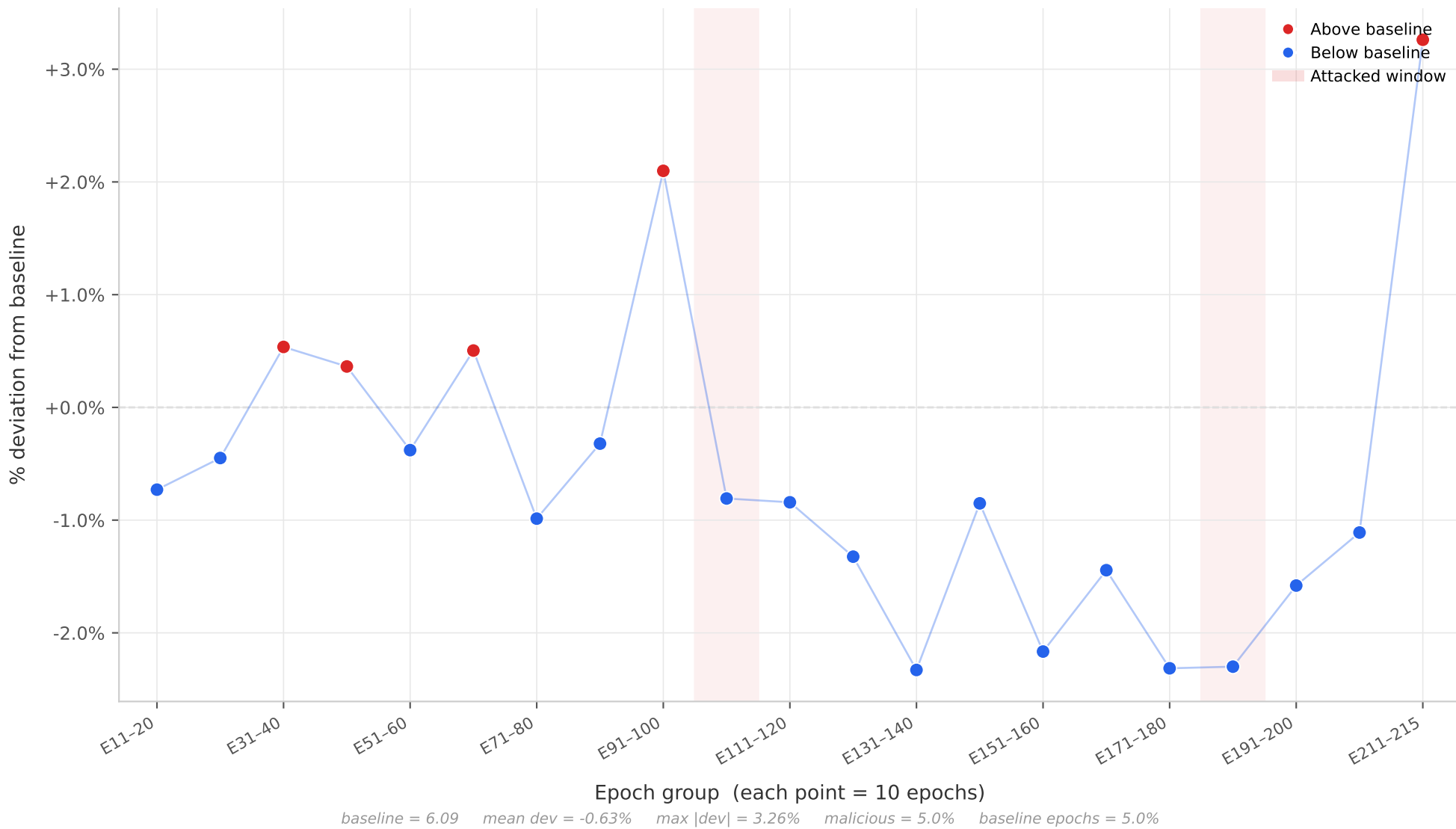
Source IP Entropy

Entropy



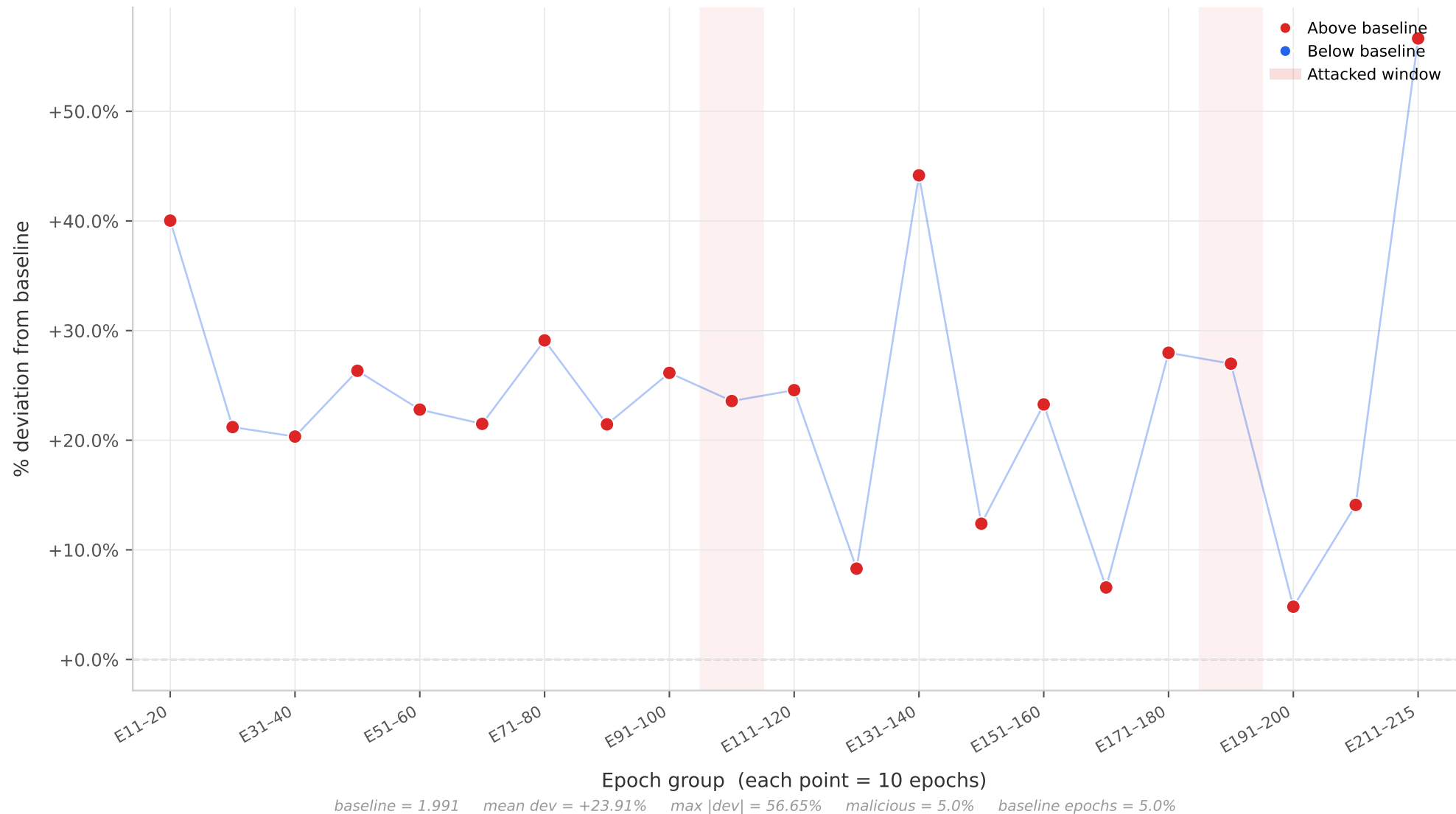
Destination IP Entropy

Entropy



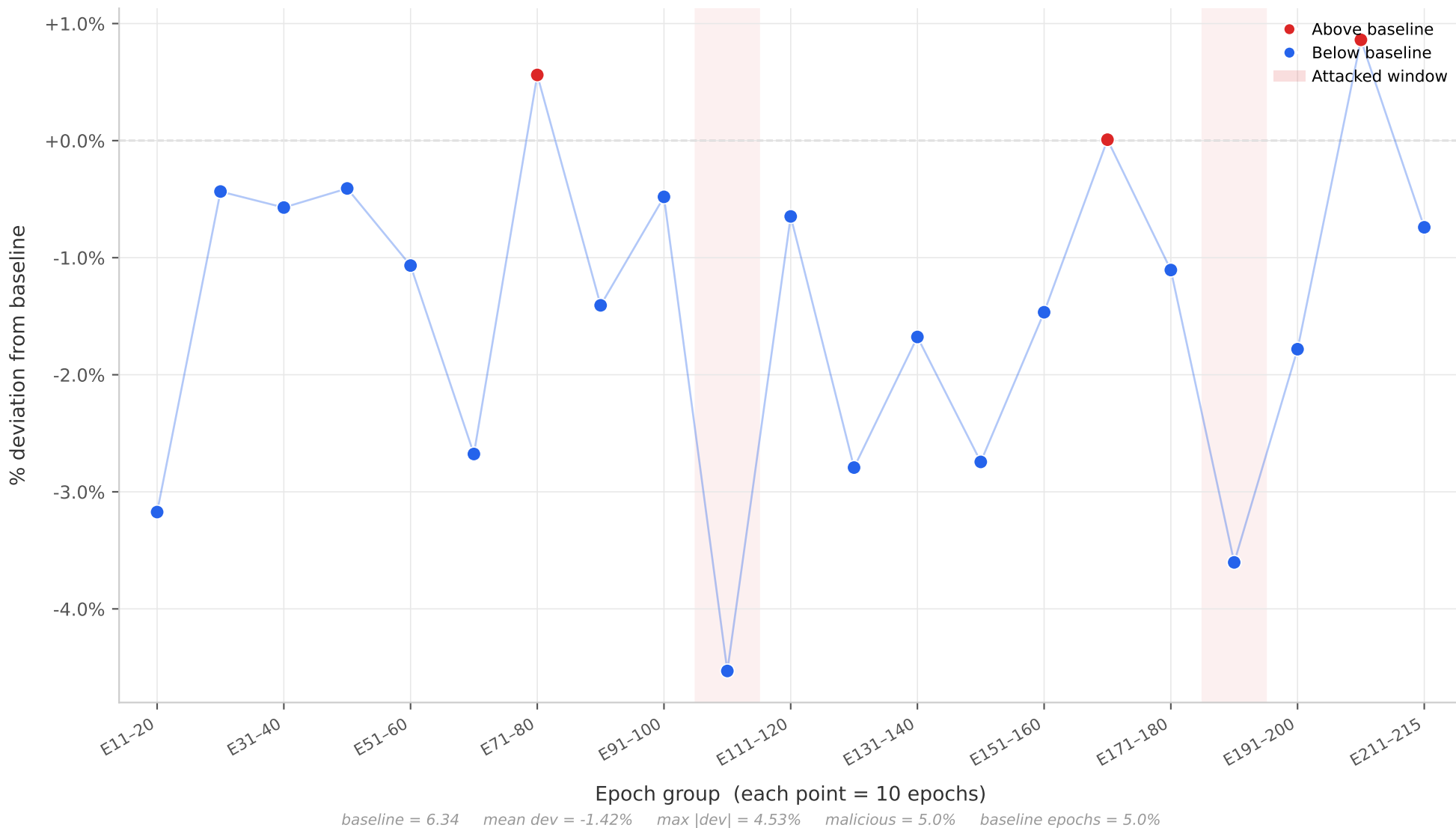
Source Port Entropy

Entropy



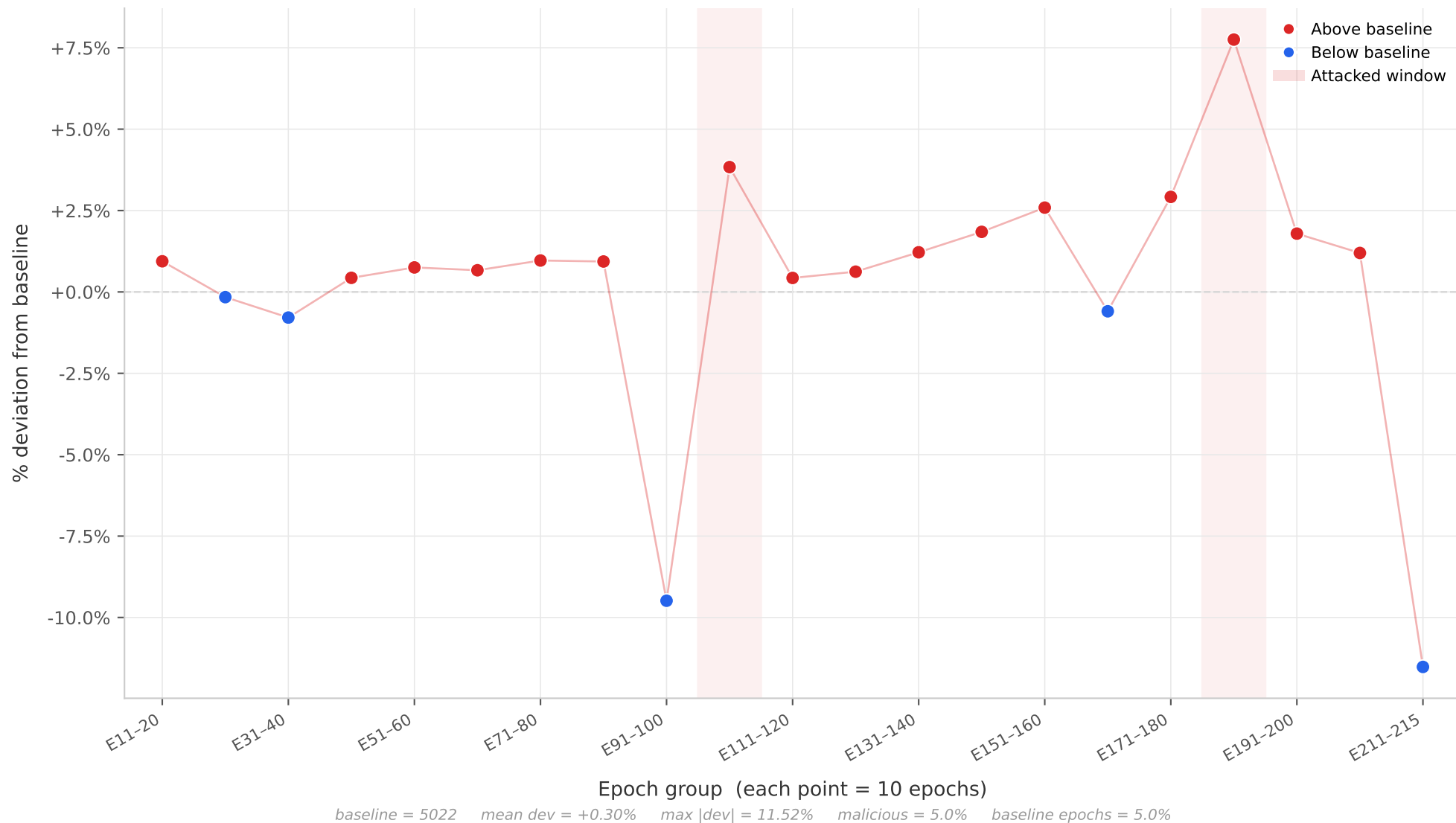
Destination Port Entropy

Entropy



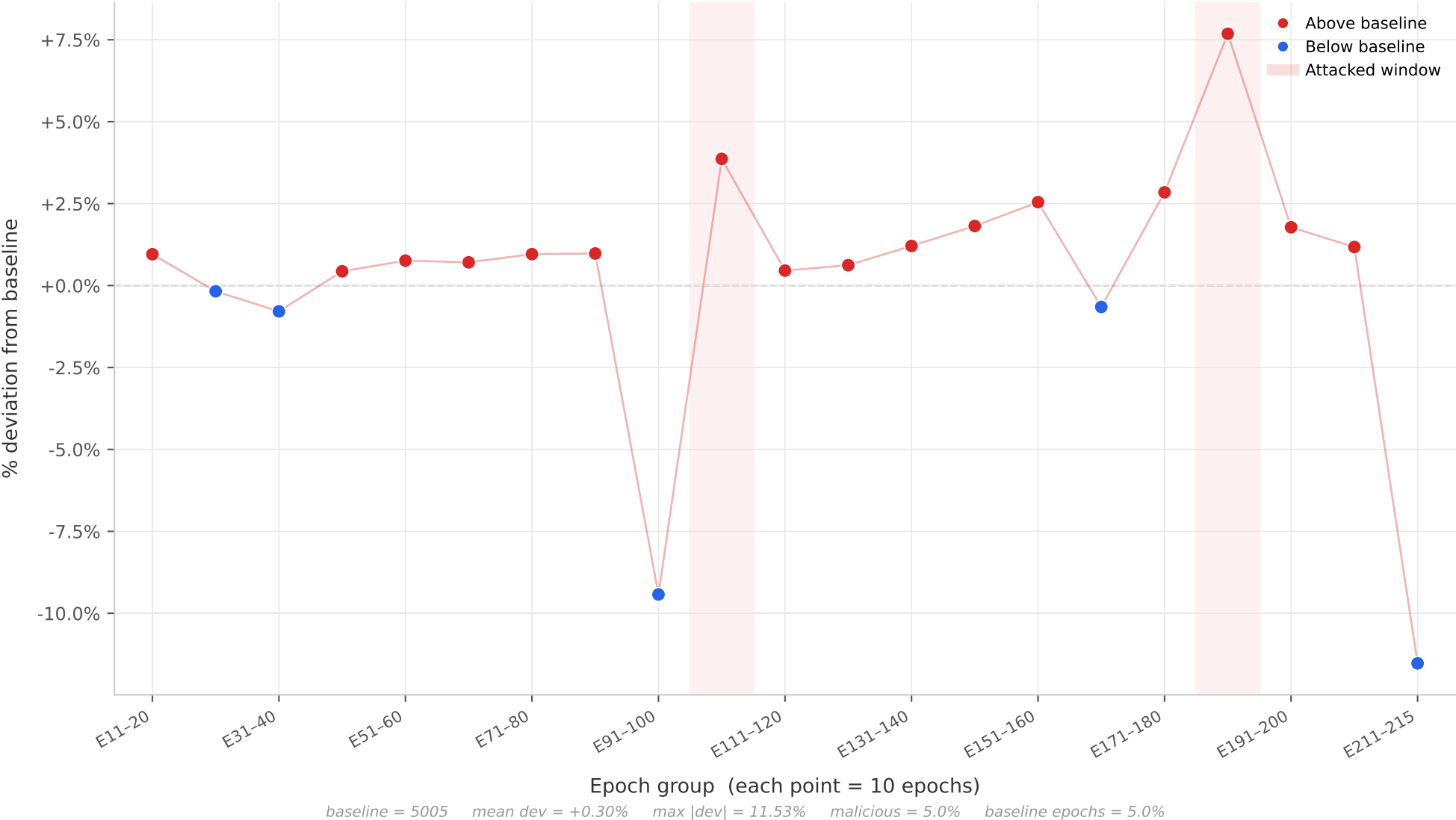
Source IP Cardinality

Cardinality



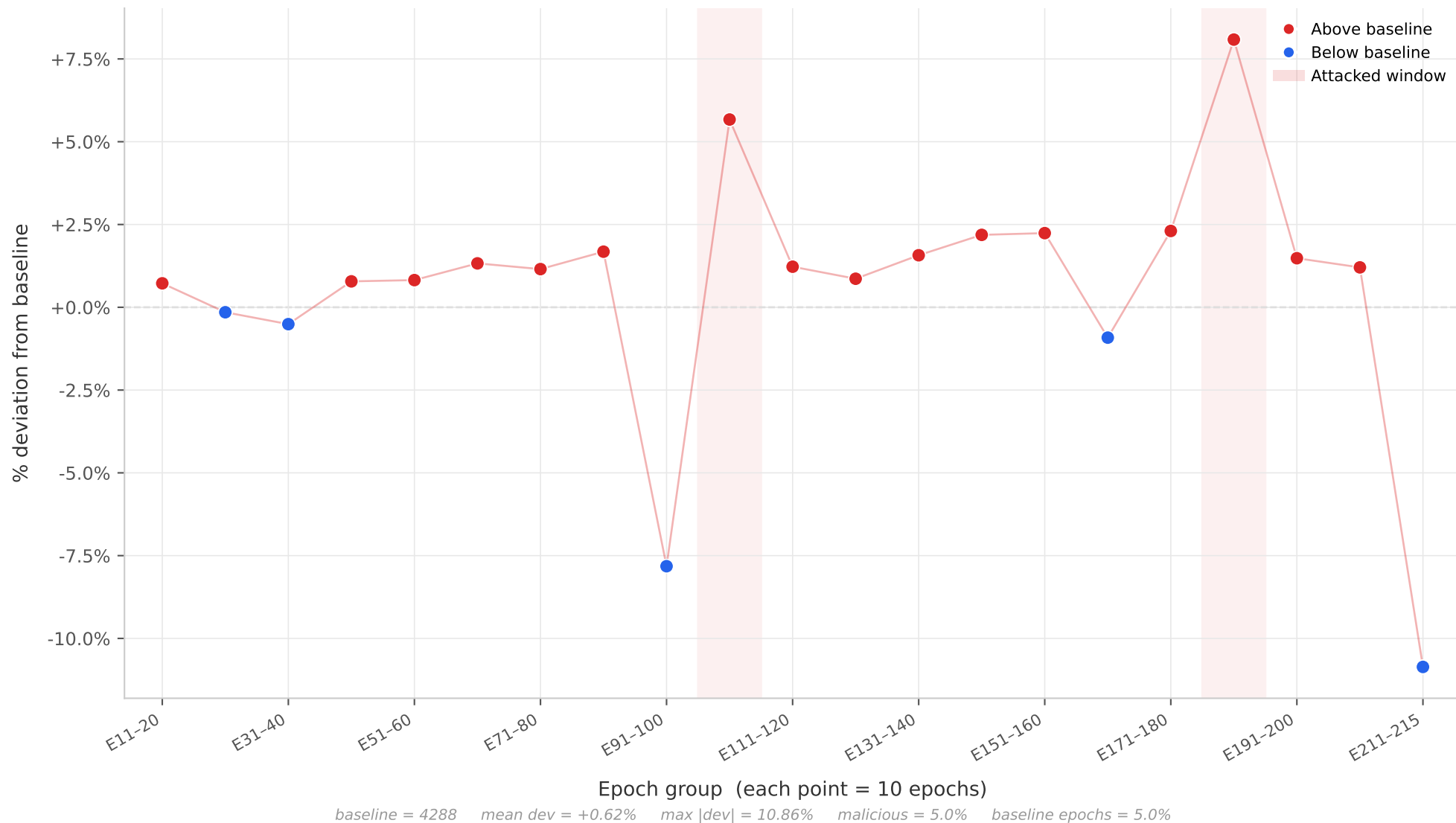
Destination IP Cardinality

Cardinality



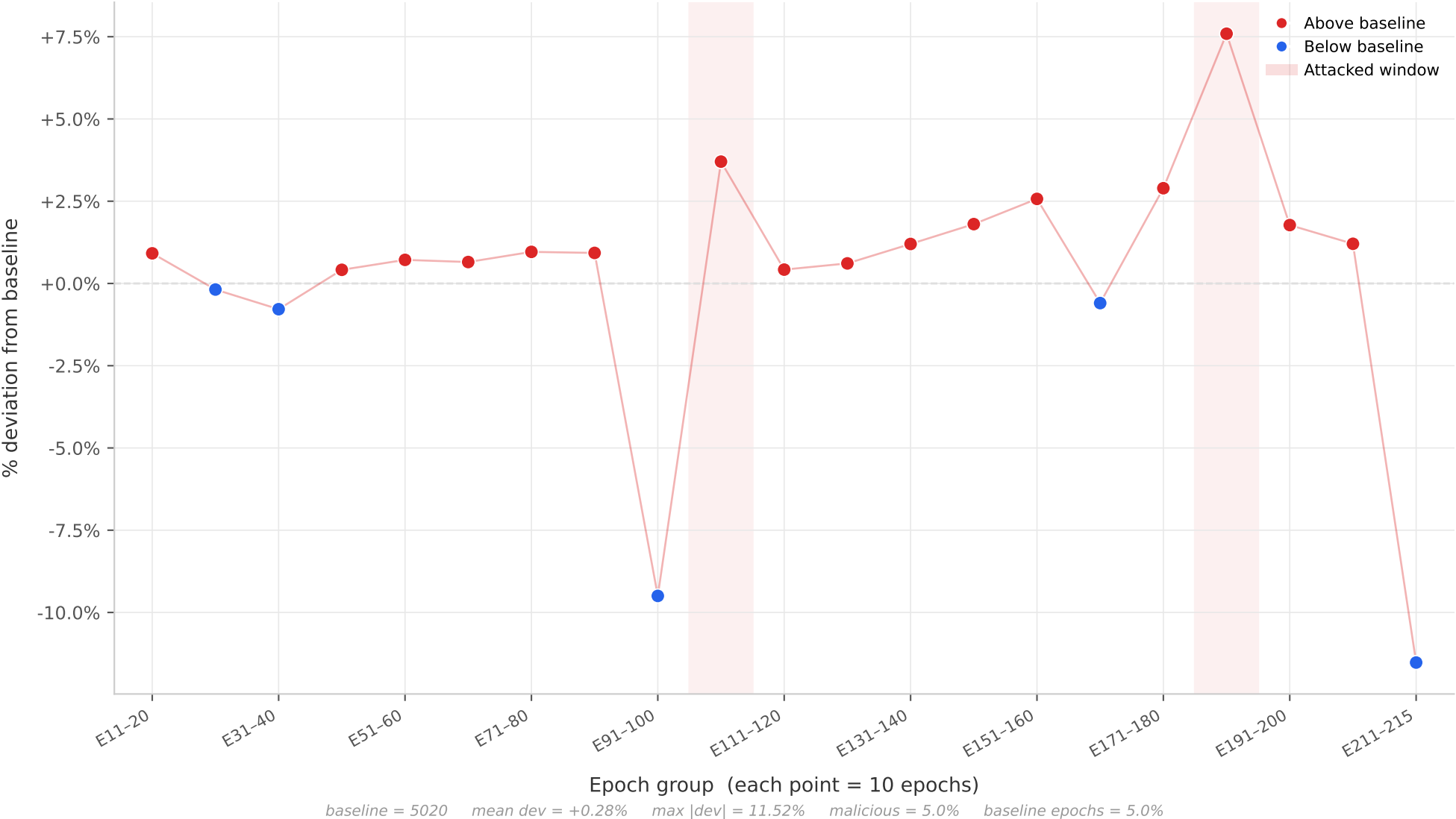
Source Port Cardinality

Cardinality



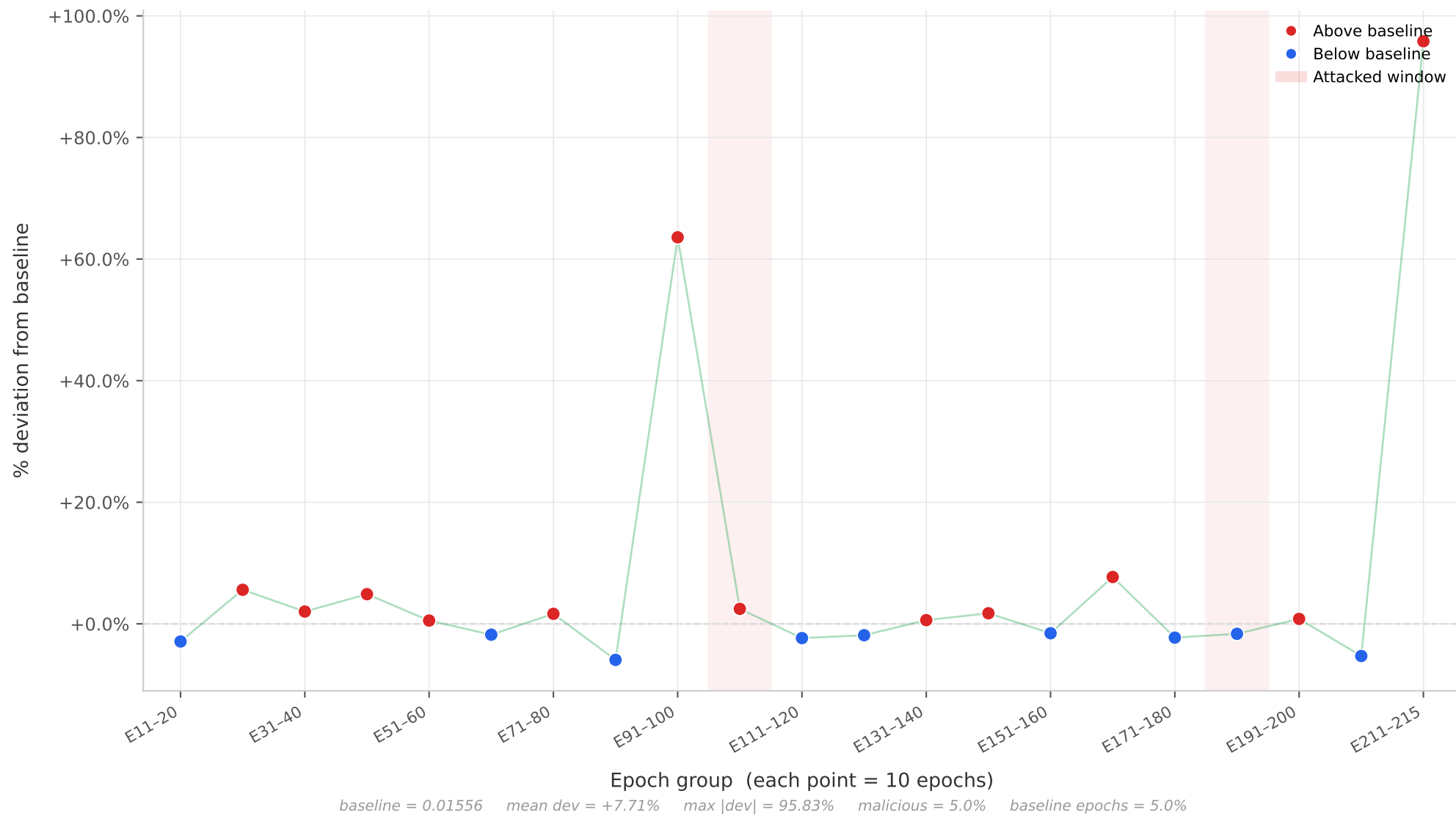
Destination Port Cardinality

Cardinality



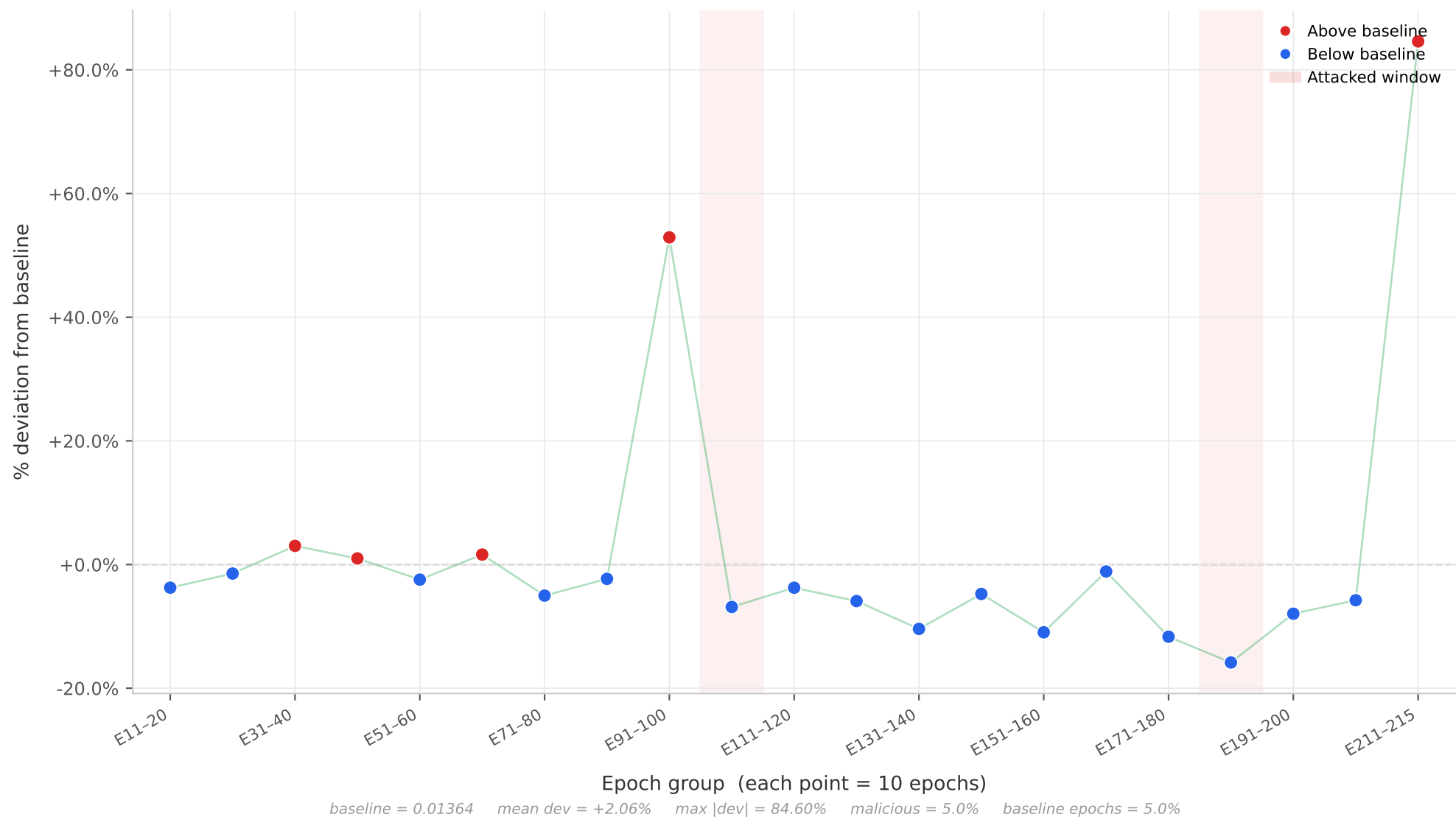
Source IP Uniformity

Uniformity



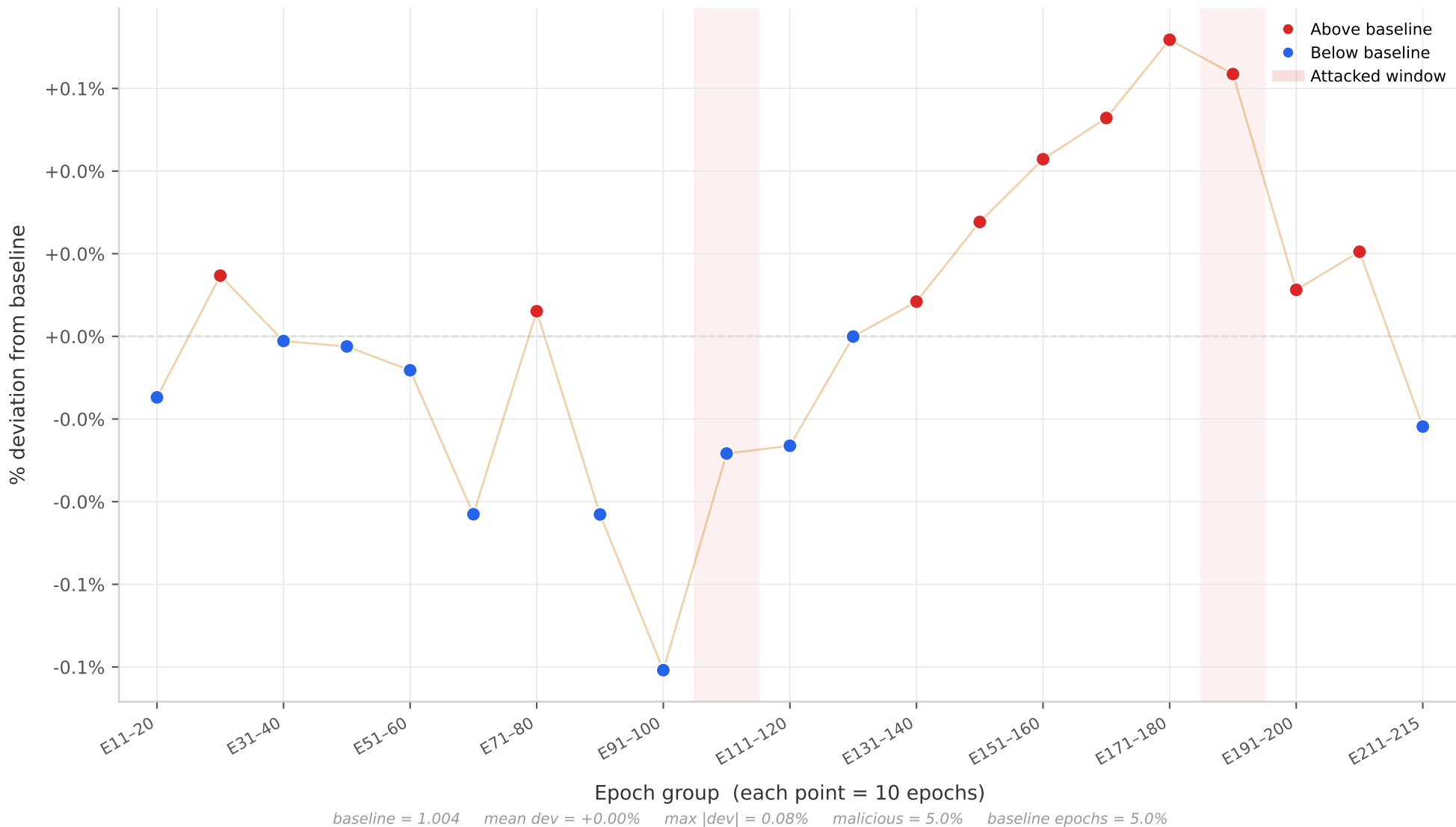
Destination IP Uniformity

Uniformity



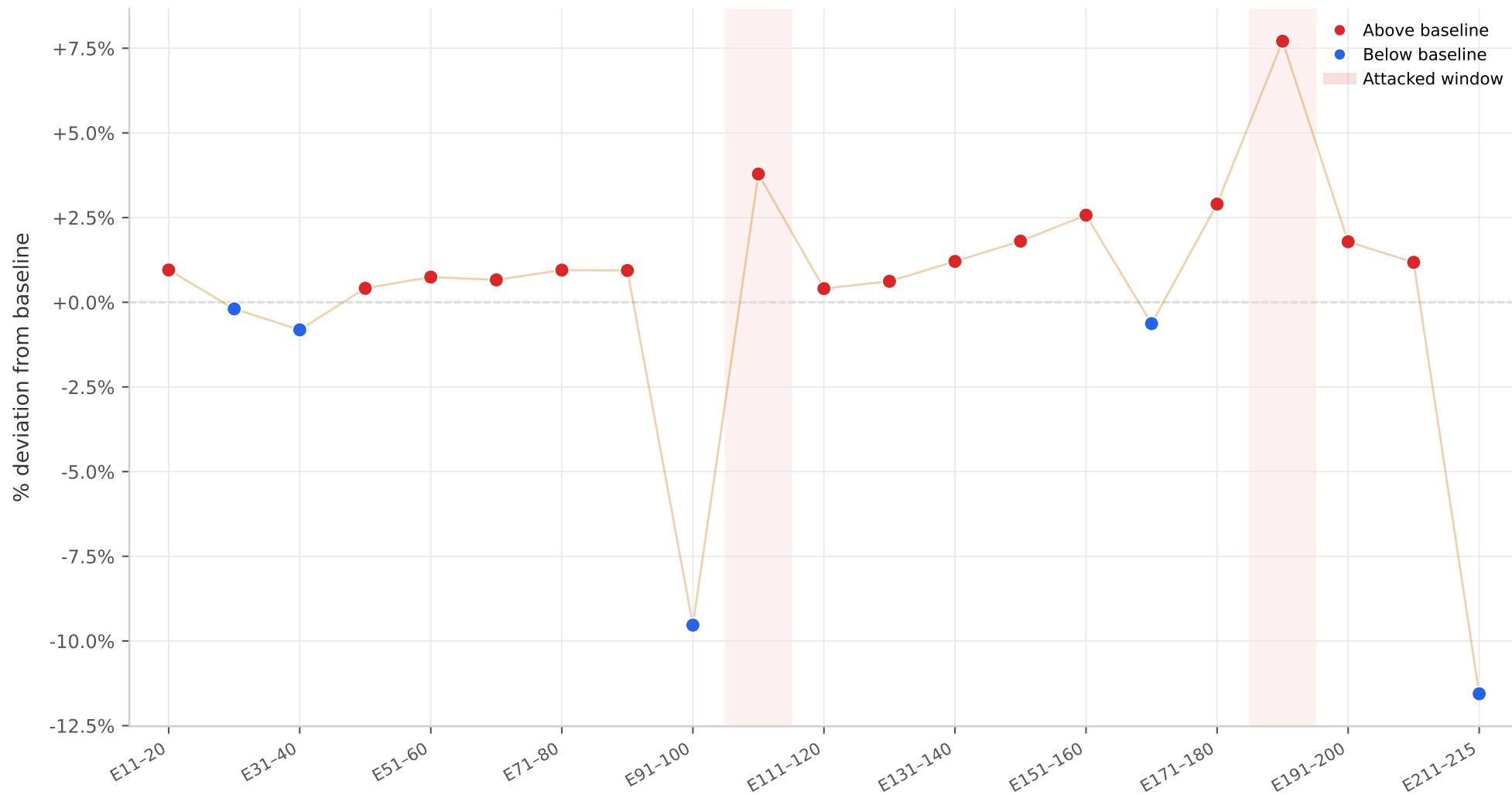
Src / Dst IP Ratio

Flow



Total Flows

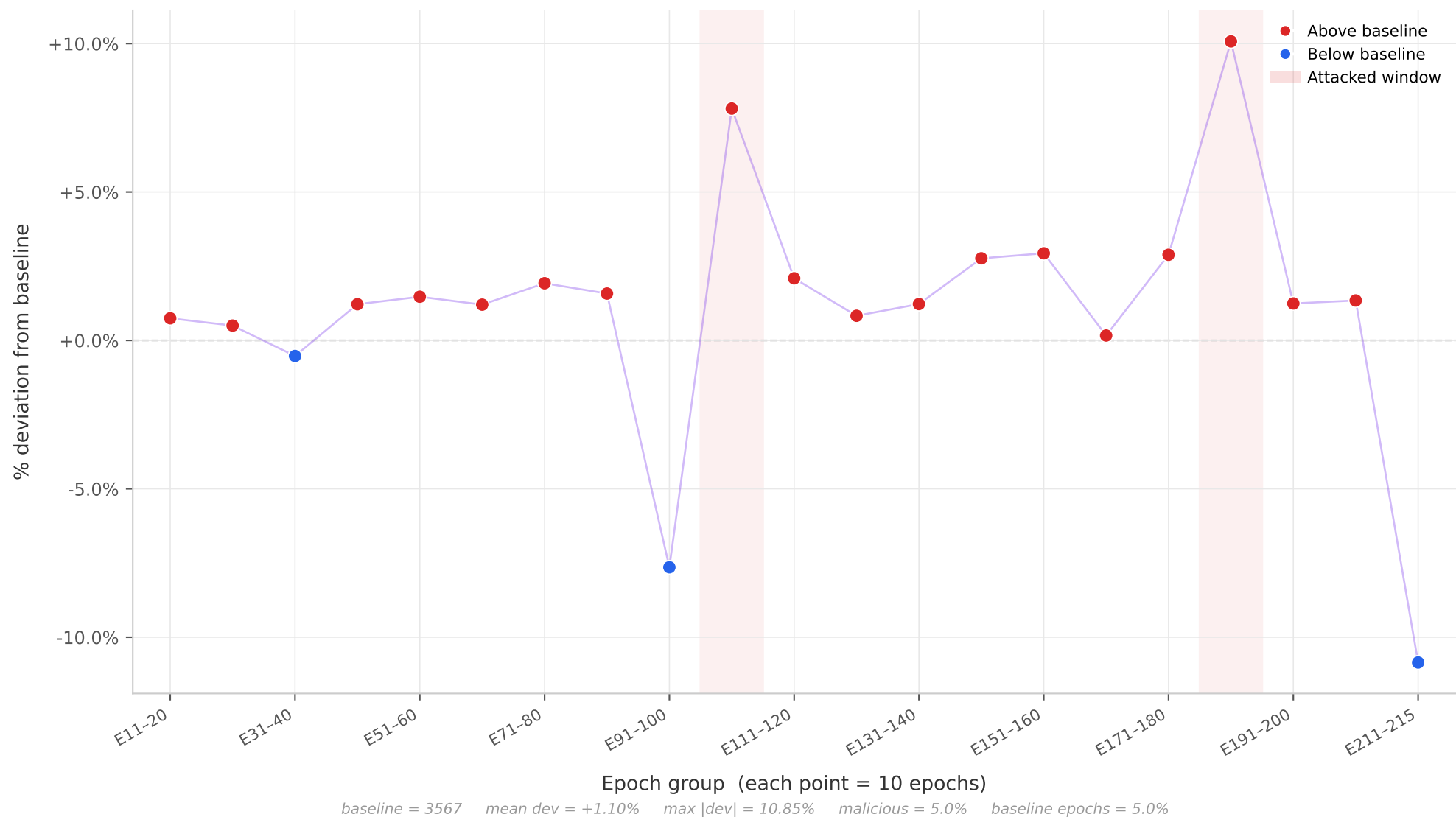
Flow



baseline = 5056 mean dev = +0.28% max |dev| = 11.56% malicious = 5.0% baseline epochs = 5.0%

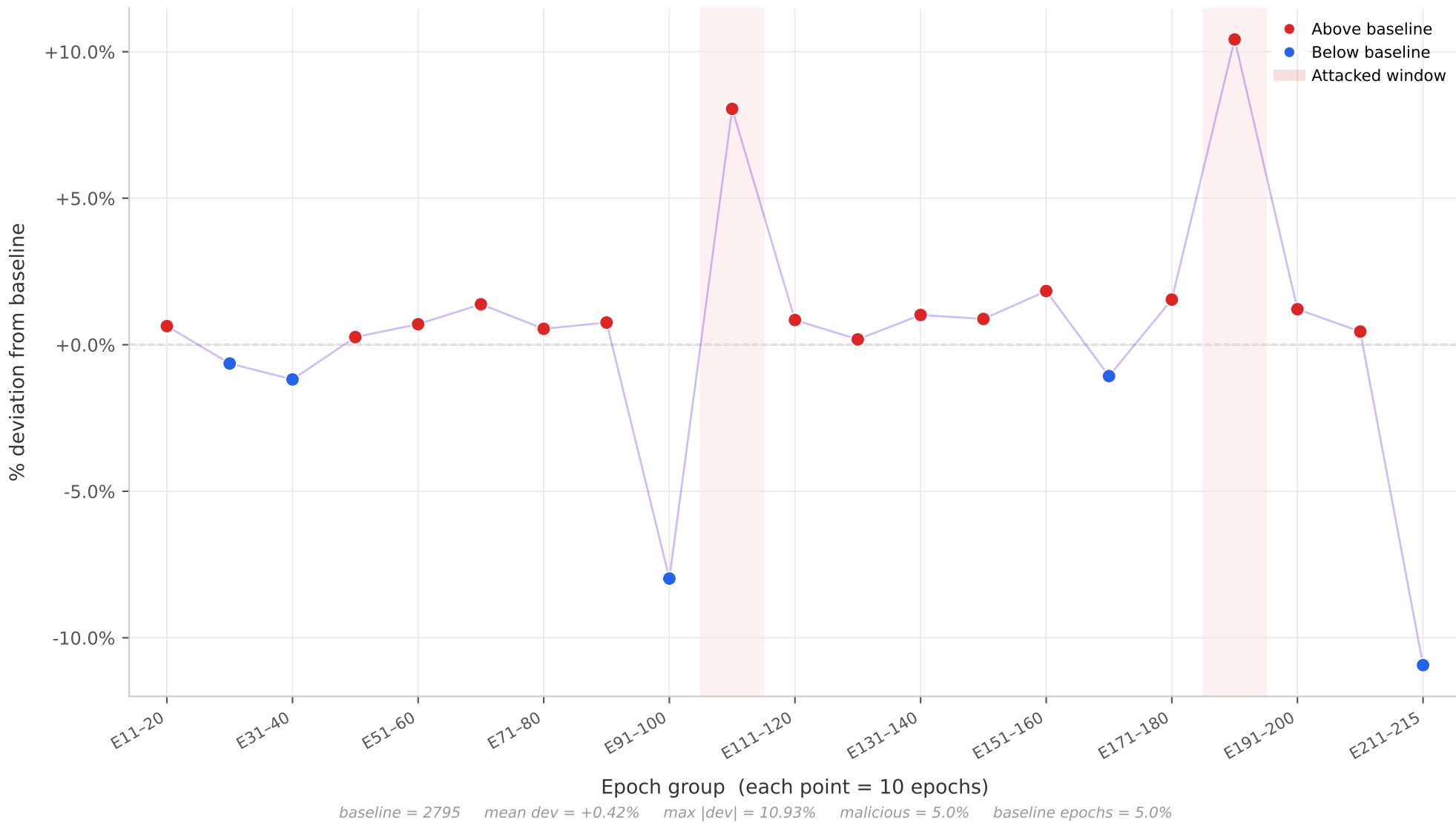
Unique Source IPs

Count



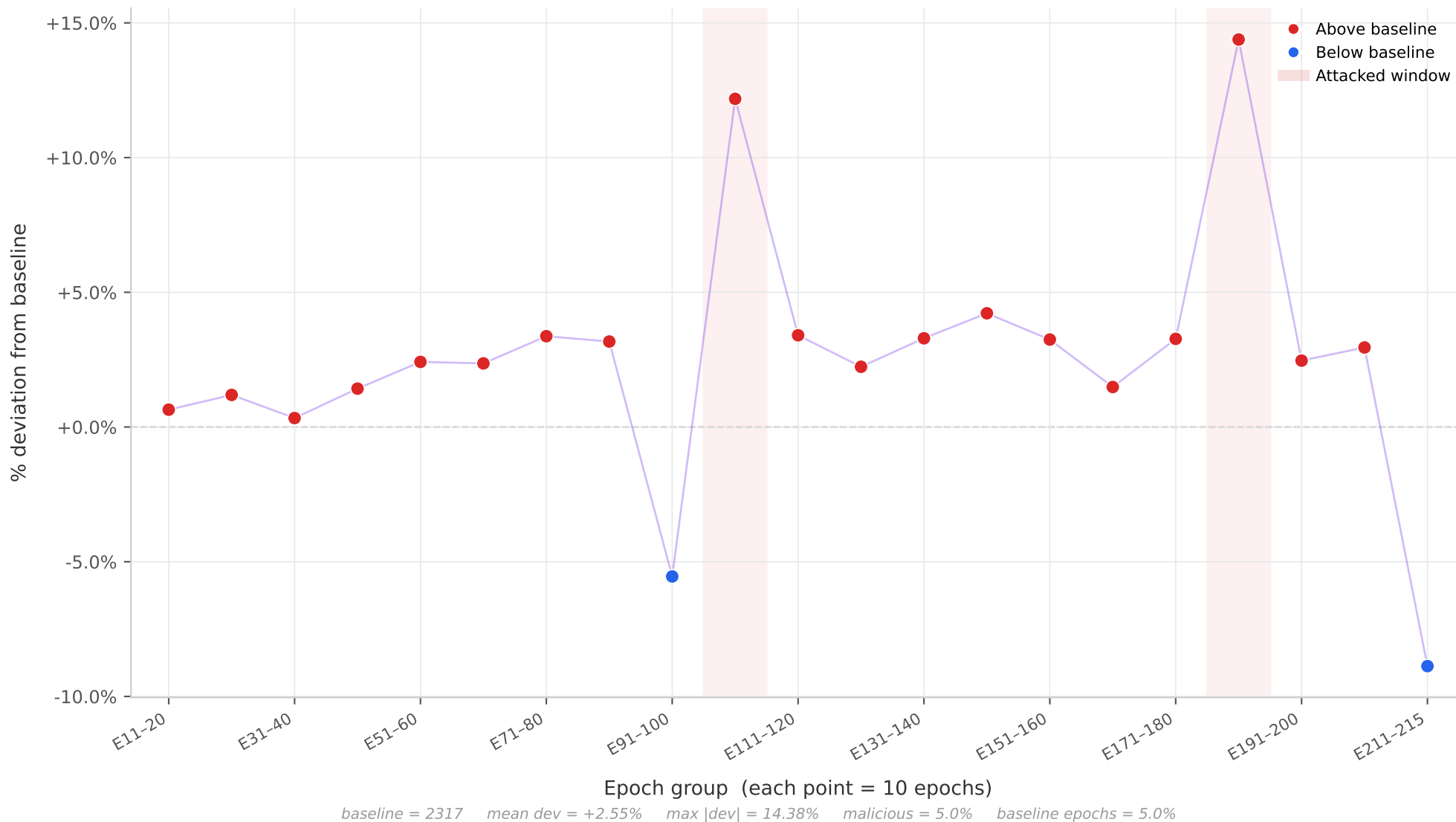
Unique Destination IPs

Count



Unique Source Ports

Count



Unique Destination Ports

Count

